

A uniform relaxation gap with nearest-neighbour interactions

For a periodic zero-field Ising heat-bath chain with per-site refresh rate $a > 0$ and dimensionless ferromagnetic coupling $b \geq 0$, the full relaxation gap is exactly $a[1 - \tanh(2b)]$ for every $N \geq 3$. This replaces G02's independence premise by a quantitative interaction contraction. Bounded b and a positive per-site clock floor give a size-independent bound; finite interaction range alone does not control joint size/coupling limits.

The exact gap is an established result, explicitly recorded by Lubetzky–Sly, preprint p. 4. B73 matches their rate-one convention to the rate a here. The following self-contained proof retains the model's role as an interacting benchmark; its foundational use separates relaxation, global mixing and physical energy normalization.

1. Model, law, clock and access

On $\Omega_N = \{-1, 1\}^N$ with cyclic indices and $N \geq 3$, let

$$\pi_{N,b}(\sigma) = Z_{N,b}^{-1} \exp \left(b \sum_{i=1}^N \sigma_i \sigma_{i+1} \right), \quad b = \beta J \geq 0.$$

Here J is an energy and β an inverse energy; b is dimensionless. Every site has an independent rate- a Poisson refresh clock. Upon a ring, replace its sign by the conditional Gibbs sign, whose plus probability is

$$p_i(\sigma) = \frac{1 + \tanh[b(\sigma_{i-1} + \sigma_{i+1})]}{2} = \frac{1}{2} + \frac{\theta}{4}(\sigma_{i-1} + \sigma_{i+1}), \quad \theta = \tanh(2b).$$

The generator on all functions on Ω_N is $Q_N = a \sum_i (E_i - I)$, where E_i conditions on all sites except i . Refreshes may leave the sign unchanged. The actual flip rate is $a[1 - \theta\sigma_i(\sigma_{i-1} + \sigma_{i+1})/2]/2$. The conditional-expectation identity makes each E_i an orthogonal projection in $L^2(\pi_{N,b})$, hence Q_N is self-adjoint and reversible. For finite b , all conditional probabilities are strictly between zero and one, so single-spin flips connect all states and the chain is irreducible. Its full centered-space gap has inverse-time units. The time is an auxiliary stochastic evolution clock, specified independently of the invariant law.

Fix a velocity calibration $u > 0$, independent of N, b , and observe $v_i = u\sigma_i$. Spin-flip symmetry centers these observables; each has variance u^2 and two readout levels separated by $2u$. The full vector identifies the configuration. Its linear span need not cover all centered functions: the proof below controls the full spectrum through the dynamics.

2. All-size lower bound by a written coupling proof

Couple two copies with the same site clocks and the same uniform random number at every refresh. Let D_i indicate disagreement at site i and $D = \sum_i D_i$ be Hamming distance. The refresh disagreement probability is $|p_i(\sigma) - p_i(\eta)|$, bounded by $\theta(D_{i-1} + D_{i+1})/2$. Thus the coupled generator satisfies

$$\mathcal{L}D \leq -aD + \frac{a\theta}{2} \sum_i (D_{i-1} + D_{i+1}) = -a(1 - \theta)D.$$

The finite-state evolution and Gronwall's inequality give $\mathbb{E}D_t \leq e^{-rt}D_0$, $r = a(1 - \theta) > 0$. For the Hamming Lipschitz seminorm of any real function f ,

$$|P_t f(\sigma) - P_t f(\eta)| \leq \text{Lip}(f) e^{-rt} d(\sigma, \eta), \quad \text{Lip}(P_t f) \leq e^{-rt} \text{Lip}(f).$$

Every nonconstant real eigenfunction of $-Q_N$ with eigenvalue λ has positive finite seminorm, and $P_t f = e^{-\lambda t} f$. Cancellation yields $e^{-\lambda t} \leq e^{-rt}$ for every $t > 0$, so $\lambda \geq r$. Reversibility supplies

a real eigenbasis. This proves the lower bound for every centered mode, including functions invisible to the linear span of the local observables. No dimension-dependent prefactor enters the spectral estimate.

3. A matching eigenmode and susceptibility (C124)

For $M = \sum_i \sigma_i$, conditional expectation gives

$$Q_N \sigma_i = -a \sigma_i + \frac{a\theta}{2}(\sigma_{i-1} + \sigma_{i+1}), \quad Q_N M = -a(1 - \theta)M.$$

M is centered and nonzero in $L^2(\pi_{N,b})$, since every configuration has positive probability. The upper bound from this eigenfunction matches §2, proving

$$\boxed{\text{gap}(-Q_N) = a[1 - \tanh(2b)] \quad (N \geq 3).}$$

For the bounded mean velocity $V = uM/N$, $|V| \leq u$ and

$$\chi(V) = \frac{\text{Var}_\pi(V)}{a[1 - \tanh(2b)]}, \quad \frac{\chi(V)}{\text{Var}_\pi(V)} = \frac{1}{\text{gap}(-Q_N)}.$$

The variance is positive for every finite member but is not asserted to have a uniform lower bound in N . A fixed mass $m > 0$ gives the action-valued plateau $H(V) = 2m\chi(V)$, as in C019/C041. The normalized response has time units and detects the slow mode without an extensive full-frame sum.

4. Premises retained and closing limits

At fixed finite b , the exact gap stays positive as $N \rightarrow \infty$; this statement concerns the sequence of finite-volume gaps. More generally, $a_N \geq a_0 > 0$ and $0 \leq b_N \leq B < \infty$ imply $\text{gap}(-Q_N) \geq a_0[1 - \tanh(2B)]$. At $b = 0$ the refresh chain flips with rate $a/2$, recovering C046's gap a .

Finite range and fixed calibration persist if $b_N \rightarrow \infty$ with a fixed, but the exact gap tends to zero. For example $b_N = (\log N)/4$ gives $\tanh(2b_N) = (N - 1)/(N + 1)$ and gap $2a/(N + 1)$. This varies interaction strength as well as size; it is not a fixed-coupling thermodynamic closure. Alternatively dividing Q_N by N fixes the total refresh rate at a and gives gap $a[1 - \tanh(2b)]/N$ even at fixed b .

The sufficient replacement for independence in this family is a uniform margin in the neighbour-influence sum, together with the per-site clock floor. A prescribed invariant Gibbs law alone selects neither clock. For a supplied action constant $K > 0$, the operator $K(-Q_N)$ has energy gap $Ka[1 - \tanh(2b)]$. This does not identify the sampling operator with a physical quantum Hamiltonian or select K . An infinite-volume operator construction and a continuum or quantum-field transfer remain separate tasks.

5. Evidence and strategic consequence

The proof is a finite-state written contraction argument plus an explicit magnetization eigenfunction. The Ising dynamics and relaxation mechanism are classical prior literature; no novelty is asserted. The B68 audit records model/method matches and the written proof review. Glauber formula images remain unverified after the publisher access failure; the exact full-spectrum equality is accepted on the self-contained proof, with no imported source theorem required.

G03's first milestone ends here: independence is unnecessary for a uniform finite-volume gap in this interacting family, while finite range without coupling and clock control is insufficient. Q01 is the selected next main track. Further Ising variants are parked until they discharge a named physical-operator or uniform-limit dependency.